

HDOC Day-21

Write a menu driven C program that accepts two positive integers, m and n, where $m \leq n$. Display the following menu and accept the user's choice. Use a switch statement to perform the selected operations:

1. Print the first 10 multiples of each integer from m to n.
2. Print the addition table of each integer from m to n, using addends from 1 to 10.
3. Print the subtraction table of each integer from m to n, using subtrahends from 1 to 10.
4. Print the results obtained by dividing each integer from m to n by the integers from 1 to 10. Display the results rounded to two decimal places.
5. Exit the program.

1. Algorithm

Step 1

Start the program.

Step 2

Accept two positive integers m and n.

Step 3

Check whether:

- $m > 0$
- $n > 0$
- $m \leq n$

If not, display an error message and terminate.

Step 4

Display the menu:

1. Multiplication / Multiples
2. Addition Table
3. Subtraction Table
4. Division Table

5. Exit

Step 5

Accept the user's choice.

Step 6

Use a switch statement:

Case 1:

For every number from m to n, print its first 10 multiples.

Case 2:

For every number from m to n, print its addition table using addends 1 to 10.

Case 3:

For every number from m to n, print its subtraction table using subtrahends 1 to 10.

Case 4:

For every number from m to n, divide it by 1 to 10 and print the answer up to two decimal places.

Case 5:

Exit the program.

Default:

Display "Invalid menu choice".

Step 7

Continue displaying the menu until the user selects option 5.

Step 8

Stop.

C Program:

```
#include <stdio.h>
int main()
{
    int m, n, choice;
    int i, j;

    printf("Enter two positive integers (m <= n): ");
    scanf("%d %d", &m, &n);

    if (m <= 0 || n <= 0 || m > n)
    {
```

```
printf("Invalid input! Please enter positive integers with m <= n.\n");  
return 0;  
}
```

```
do  
{  
    printf("\n===== MENU =====\n");  
    printf("1. Print first 10 multiples\n");  
    printf("2. Print addition table\n");  
    printf("3. Print subtraction table\n");  
    printf("4. Print division results\n");  
    printf("5. Exit\n");  
    printf("===== \n");
```

```
    printf("Enter your choice: ");  
    scanf("%d", &choice);
```

```
    switch (choice)  
    {  
        case 1:  
            printf("\n--- First 10 Multiples ---\n");  
  
            for (i = m; i <= n; i++)  
            {  
                printf("\nMultiples of %d:\n", i);  
  
                for (j = 1; j <= 10; j++)  
                {  
                    printf("%d x %d = %d\n", i, j, i * j);  
                }  
            }  
            break;
```

```
        case 2:  
            printf("\n--- Addition Tables ---\n");  
  
            for (i = m; i <= n; i++)  
            {  
                printf("\nAddition table of %d:\n", i);  
  
                for (j = 1; j <= 10; j++)
```

```
    {  
        printf("%d + %d = %d\n", i, j, i + j);  
    }  
}  
break;
```

case 3:

```
printf("\n--- Subtraction Tables ---\n");  
  
for (i = m; i <= n; i++)  
{  
    printf("\nSubtraction table of %d:\n", i);  
  
    for (j = 1; j <= 10; j++)  
    {  
        printf("%d - %d = %d\n", i, j, i - j);  
    }  
}  
break;
```

case 4:

```
printf("\n--- Division Results ---\n");  
  
for (i = m; i <= n; i++)  
{  
    printf("\nDivision results for %d:\n", i);  
  
    for (j = 1; j <= 10; j++)  
    {  
        printf("%d / %d = %.2f\n",  
            i, j, (float)i / j);  
    }  
}  
break;
```

case 5:

```
printf("\nExiting the program...\n");  
break;
```

default:

```

        printf("\nInvalid menu choice! Please enter a choice between 1 and
5.\n");
    }

    } while (choice != 5);

    return 0;
}

```

Test-Case Table:

Test Case	Input m,n	Choice	Expected Result
1	2,3	1	First 10 multiples of 2 and 3 are displayed
2	2,3	2	Addition tables of 2 and 3 from 1 to 10 displayed
3	2,3	3	Subtraction tables of 2 and 3 from 1 to 10 are displayed
4	2,3	4	Division results for 2 and 3 by 1-10 are displayed to 2 decimal places.
5	2,3	6	"Invalid menu choice" is displayed.
6	5,5	1	First 10 multiples of 5 are displayed.
7	0,5	-	Invalid input message is displayed.
8	5,2	-	Invalid input because $m > n$
9	-2,5	-	Invalid input message is displayed
10	2,3	5	Program exits

Compilation and Execution:

The following program has been compiled and executed successfully.

Test Case 1:

```
———— First  
Multiples of 2:  
2 * 2 = 4  
2 * 3 = 6  
2 * 4 = 8  
2 * 5 = 10  
2 * 6 = 12  
2 * 7 = 14  
2 * 8 = 16  
2 * 9 = 18  
2 * 10 = 20  
  
Multiples of 3:  
3 * 3 = 9  
3 * 4 = 12  
3 * 5 = 15  
3 * 6 = 18  
3 * 7 = 21  
3 * 8 = 24  
3 * 9 = 27  
3 * 10 = 30
```

Test Case 2:

—— Addition Tables ——

Addition table of 2:

$2 + 1 = 3$
 $2 + 2 = 4$
 $2 + 3 = 5$
 $2 + 4 = 6$
 $2 + 5 = 7$
 $2 + 6 = 8$
 $2 + 7 = 9$
 $2 + 8 = 10$
 $2 + 9 = 11$
 $2 + 10 = 12$

Addition table of 3:

$3 + 1 = 4$
 $3 + 2 = 5$
 $3 + 3 = 6$
 $3 + 4 = 7$
 $3 + 5 = 8$
 $3 + 6 = 9$
 $3 + 7 = 10$
 $3 + 8 = 11$
 $3 + 9 = 12$
 $3 + 10 = 13$

Test Case 3:

Subtraction table of 2:

$2 - 1 = 1$
 $2 - 2 = 0$
 $2 - 3 = -1$
 $2 - 4 = -2$
 $2 - 5 = -3$
 $2 - 6 = -4$
 $2 - 7 = -5$
 $2 - 8 = -6$
 $2 - 9 = -7$
 $2 - 10 = -8$

Subtraction table of 3:

$3 - 1 = 2$
 $3 - 2 = 1$
 $3 - 3 = 0$
 $3 - 4 = -1$
 $3 - 5 = -2$
 $3 - 6 = -3$
 $3 - 7 = -4$
 $3 - 8 = -5$
 $3 - 9 = -6$
 $3 - 10 = -7$

Test Case 4:

Division results for 2:

2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
2 / 4 = 0.50
2 / 5 = 0.40
2 / 6 = 0.33
2 / 7 = 0.29
2 / 8 = 0.25
2 / 9 = 0.22
2 / 10 = 0.20

Division results for 3:

3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
3 / 4 = 0.75
3 / 5 = 0.60
3 / 6 = 0.50
3 / 7 = 0.43
3 / 8 = 0.38
3 / 9 = 0.33
3 / 10 = 0.30

Test Case 5:

```
_____ MENU _____  
1. Print first 10 multiples  
2. Print addition table  
3. Print subtraction table  
4. Print division results  
5. Exit
```

Enter your choice: 6

Invalid menu choice! Please enter a choice between 1 and 5